

09 Orthogonality

9.5 QR Decomposition

Decomposing any real matrix into an orthogonal matrix times an upper-triangular matrix — the matrix form of Gram–Schmidt orthogonalization's result



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

QR decomposition: decomposing any real matrix into an orthogonal matrix times an upper-triangular matrix

2

The connection between Gram–Schmidt orthogonalization and QR decomposition

3

Distinguishing complete QR decomposition from reduced (economic) QR decomposition

4

Every column of upper-triangular R holds the original column's coordinates in the orthogonal basis

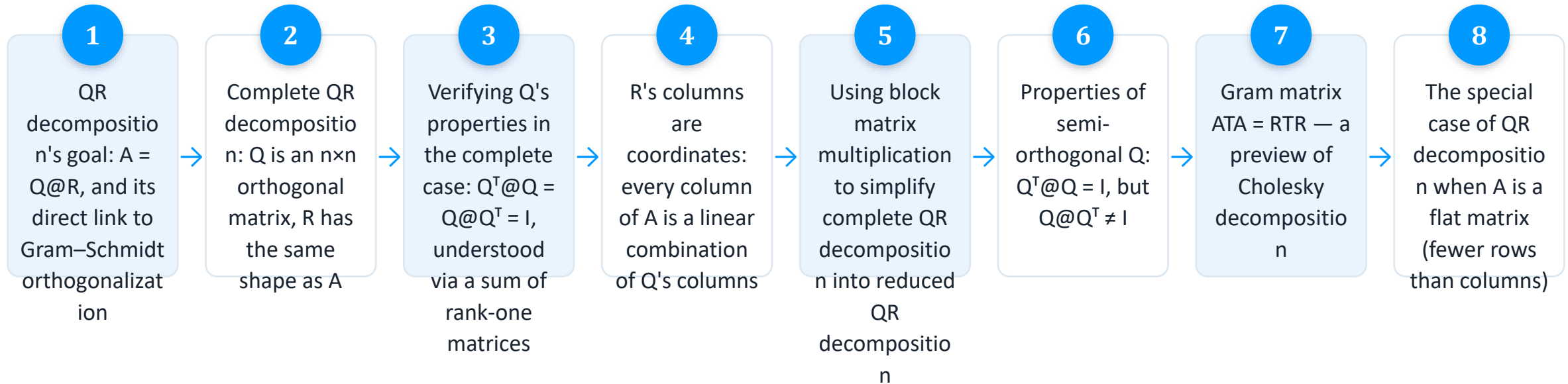
5

The relationship between a matrix's QR decomposition and its Gram matrix's Cholesky decomposition

6

What's different about QR decomposition when A is a flat matrix (fewer rows than columns)

From Complete to Reduced, Then to the Properties of Semi-Orthogonal Matrices



Seven Lenses for Every Topic in This Chapter

- 1 Geometric view: the identity, rotation, reflection, permutation matrices, and their combinations are all orthogonal
- 2 Algebraic property: $\mathbf{Q}^T = \mathbf{Q}^{-1}$ — transpose equals inverse; the determinant is always +1 or -1
- 3 Gram-matrix view: the columns are pairwise orthogonal, and each column is a unit vector
- 4 Projection view: using an orthogonal matrix's columns to project — the sum of projections restores the vector
- 5 Orthogonal-complement view: every subspace has an orthogonal complement perpendicular to it everywhere
- 6 How Gram–Schmidt orthogonalization and QR decomposition build an orthonormal basis from any basis
- 7 Where orthogonal matrices show up: regression, spectral decomposition, SVD, principal component analysis

QR decomposition's goal is to split a matrix into an orthogonal matrix times an upper-triangular matrix

$$\mathbf{A} = \mathbf{Q}\mathbf{R} \quad (\mathbf{Q} \text{ is orthogonal, } \mathbf{R} \text{ is upper triangular})$$

- Gram–Schmidt orthogonalization takes full-column-rank $\mathbf{A}$'s columns as input, orthogonalizes them column by column, and the resulting pairwise-orthogonal unit vectors are exactly $\mathbf{Q}$'s columns
- Along the way, each original column's projection coefficients onto these orthogonal directions form matrix $\mathbf{R}$ — classical Gram–Schmidt can be numerically unstable, so practical implementations often use Householder transformations and similar methods instead

QR Decomposition Comes in Two Common Forms

Complete: $\mathbf{Q}$ is an orthogonal matrix (square); any real matrix admits a complete QR decomposition

Reduced (thin): $\mathbf{Q}$ is a semi-orthogonal matrix, with the same shape as the original matrix

Figure 1 · Complete QR Decomposition

- $A_{n \times p} = Q_{n \times n} @ R_{n \times p}$ — Q is an $n \times n$ orthogonal matrix, R has the same shape as A

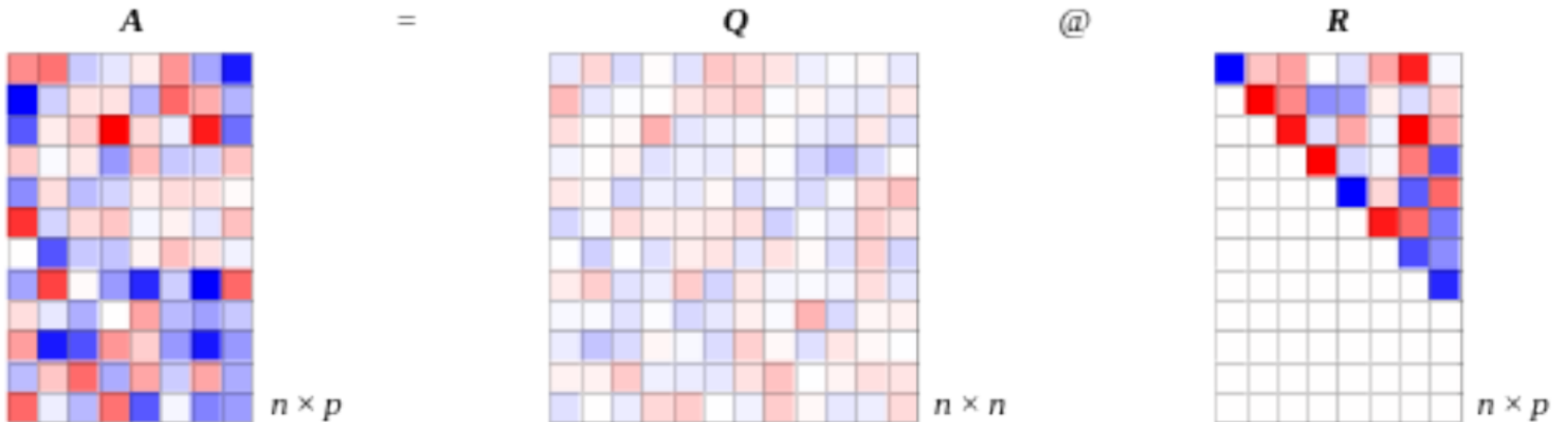
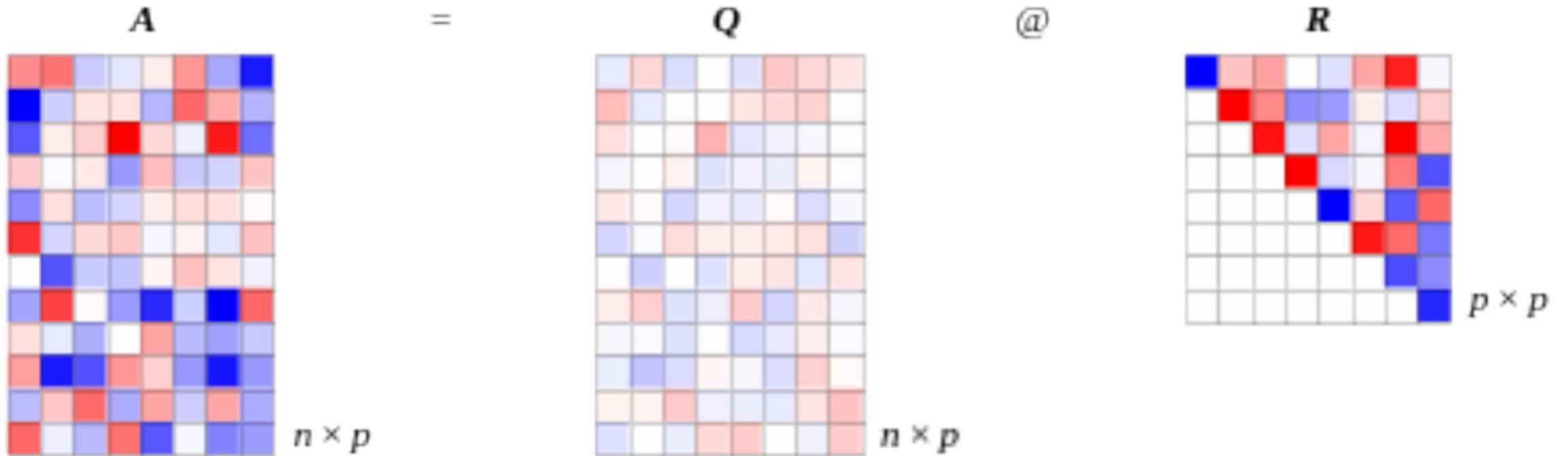


Figure 2 · Reduced QR Decomposition

- Q 's columns form an orthonormal basis when A has full column rank



In complete QR decomposition, the square matrix Q is orthogonal

$$Q_{n \times n} @ Q_{n \times n}^T = Q_{n \times n}^T @ Q_{n \times n} = I_{n \times n}$$

- Q 's columns $[q_1, q_2, \dots, q_n]$ form an orthonormal basis spanning $\mathbb{R}^n$

Figure 3 · Heatmap of $Q^T Q = I$, Complete QR Decomposition

- Each entry is an inner product of two columns of Q — 1 on the diagonal, 0 elsewhere

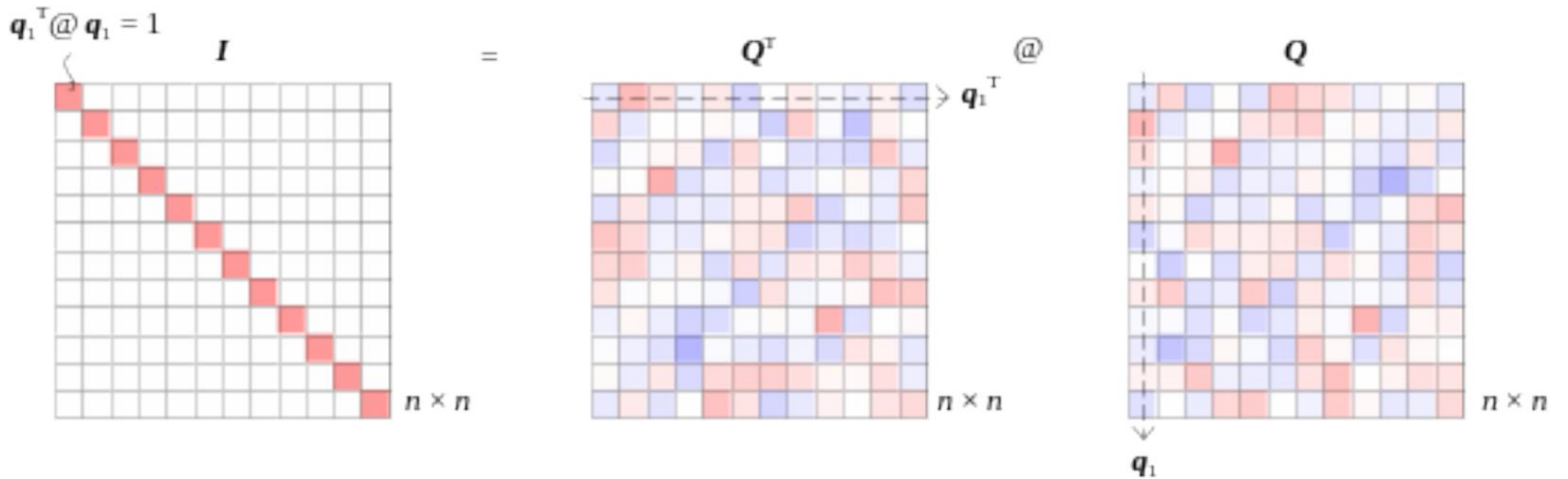


Figure 4 · $Q^T @ Q = I$, Matrix Multiplication's First View Expanded

- $q_i^T @ q_j = 1$ when $i=j$ (unit-vector self inner product), 0 when $i \neq j$ (orthogonal)

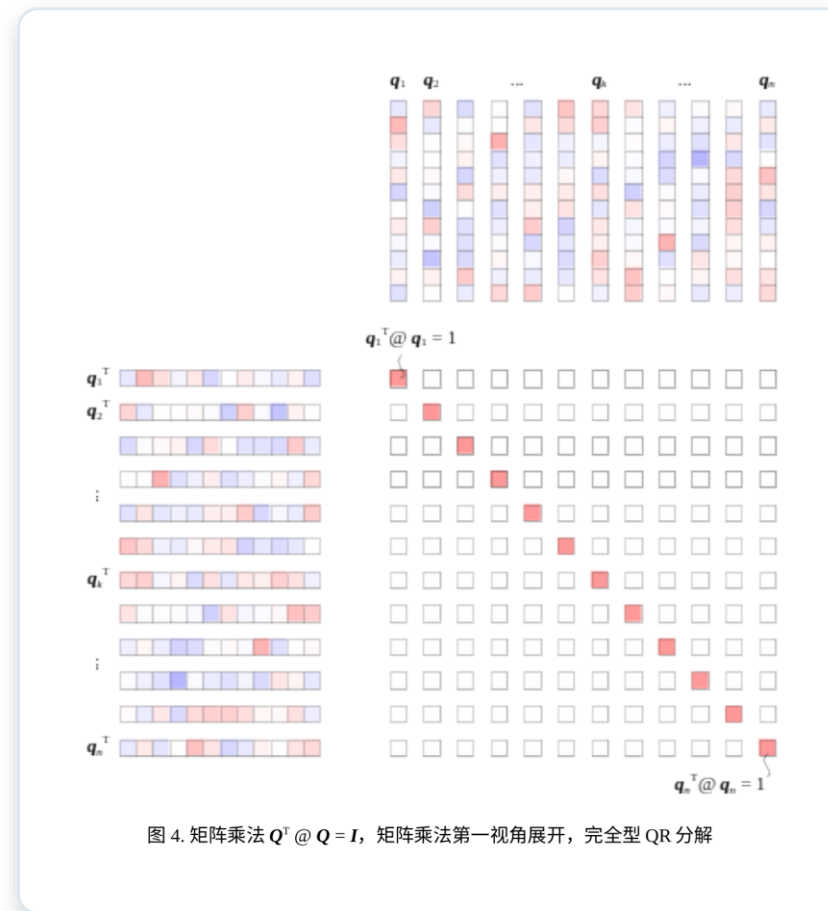
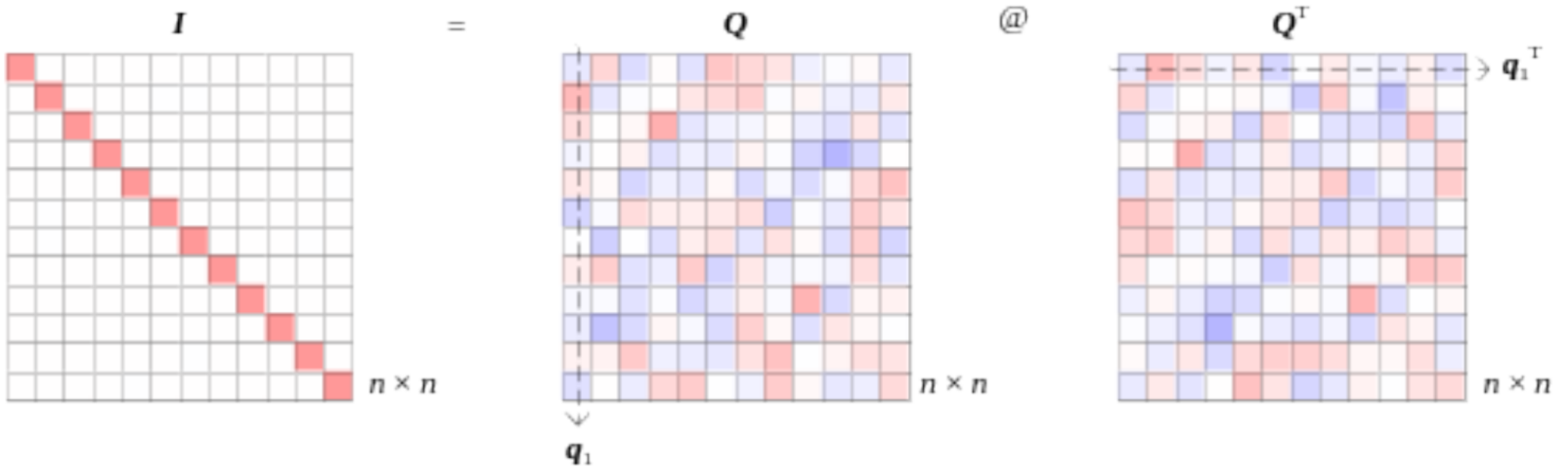


Figure 5 · Heatmap of $Q@QT=I$, Complete QR Decomposition

- Q 's rows are also pairwise-orthogonal unit vectors — another orthonormal basis



Expand $Q@Q^T=I$ using matrix multiplication's second view

$$Q@Q^T = q_1@q_1^T + q_2@q_2^T + \cdots + q_n@q_n^T = I$$

- Since Q 's columns are all nonzero vectors, each $q_i@q_i^T$ is a rank-one matrix and also symmetric

Figure 6 · Heatmap of $q_1 @ q_1^T$, Complete QR Decomposition

- $q_1 @ q_1^T$ is the projection matrix onto direction q_1 — the component q_1 contributes

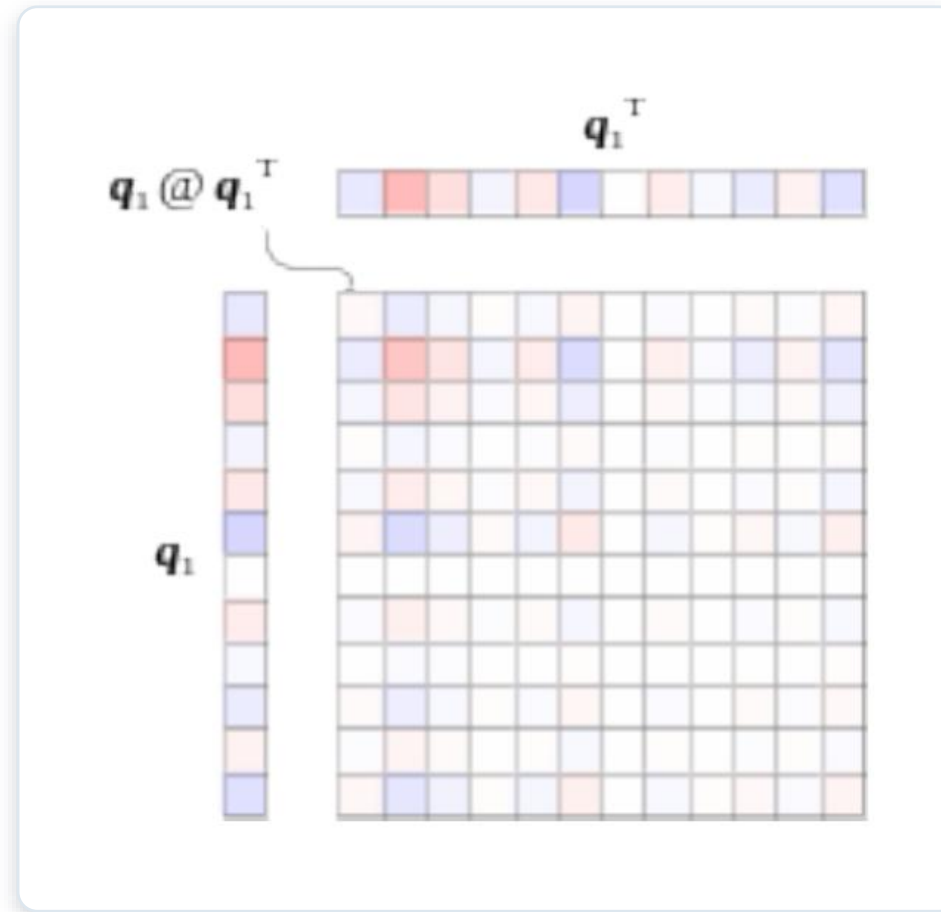


Figure 7 · n Rank-One Matrices Summing to the Identity, Complete QR Decomposition

- Each $\mathbf{q}_i @ \mathbf{q}_i^T$ projects onto a single direction; summed together, they cover all of $\mathbb{R}^n$



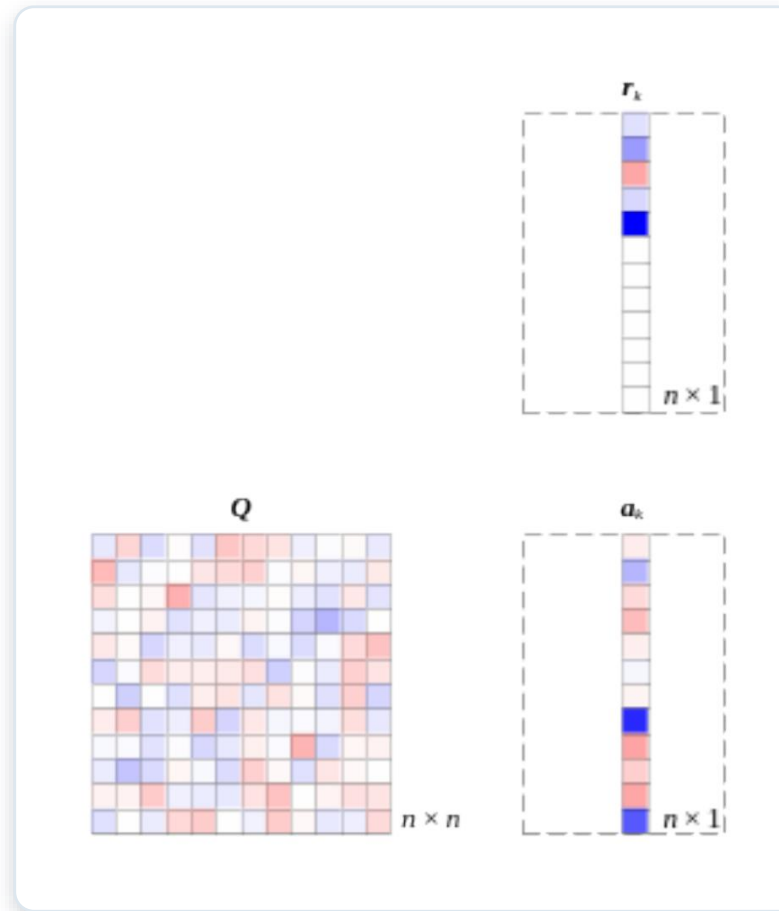
Analyzing $A = QR$ through matrix multiplication's third view — linear combination of columns

$$A = [a_1, a_2, \dots, a_p] = Q @ R = Q @ [r_1, r_2, \dots, r_p] = [Q @ r_1, Q @ r_2, \dots, Q @ r_p]$$

- A 's column a_k can be written as a linear combination of Q 's columns: $a_k = Q @ r_k = r_{1,k}q_1 + r_{2,k}q_2 + \dots + r_{n,k}q_n$

Figure 8 · Computing $\mathbf{a}_k$ via the Matrix Multiplication $\mathbf{a}_k = \mathbf{Q} @ \mathbf{r}_k$

- Multiplying column vector $\mathbf{r}_k$ by $\mathbf{Q}$ gives $\mathbf{A}$'s k -th column $\mathbf{a}_k$



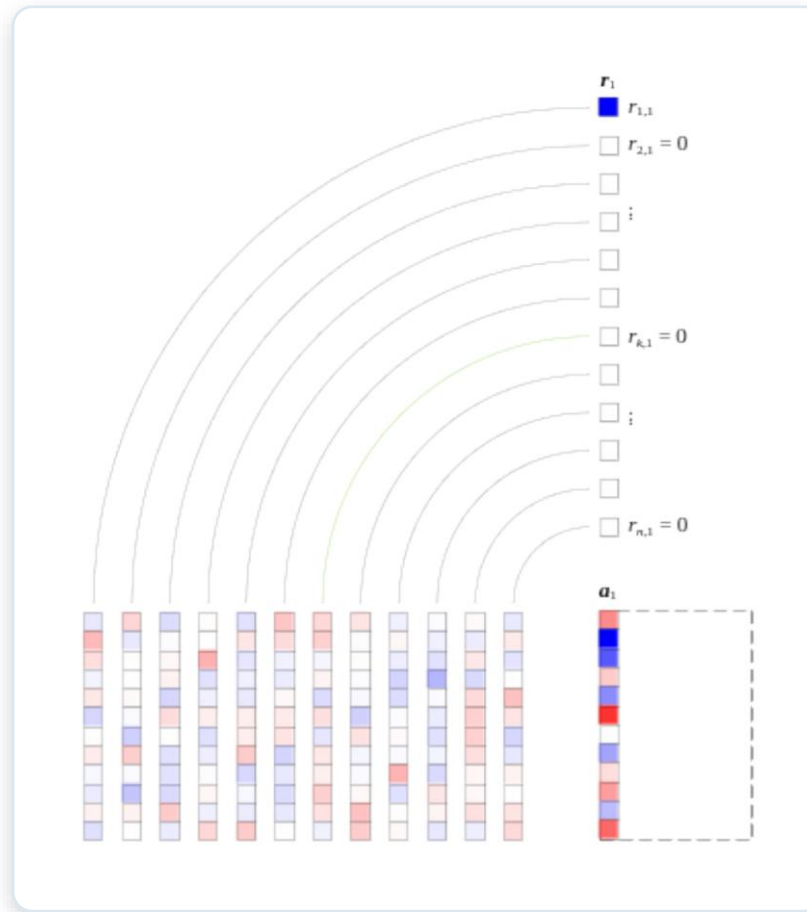
The first column $\mathbf{a}_1$ can be obtained from

$$\mathbf{a}_1 = Q\mathbf{r}_1 = r_{1,1}\mathbf{q}_1 \quad r_{1,1} = \pm\|\mathbf{a}_1\|$$

- This means $\mathbf{a}_1$'s coordinates in orthonormal basis $[\mathbf{q}_1, \dots, \mathbf{q}_n]$ are $(r_{1,1}, 0, \dots, 0)$ — $\mathbf{a}_1$ is parallel to $\mathbf{q}_1$
- $r_{1,1} = \pm\|\mathbf{a}_1\|$ shows QR decomposition is not unique; but if $\mathbf{A}$ has full column rank and $\mathbf{R}$'s diagonal entries are required to be positive, the decomposition becomes unique

Figure 9 · The Matrix Multiplication $a_1 = Q @ r_1$ for A's First Column

- Only r_1 's first entry $r_{1,1}$ is nonzero, the rest are all 0 — matching Gram–Schmidt's first step



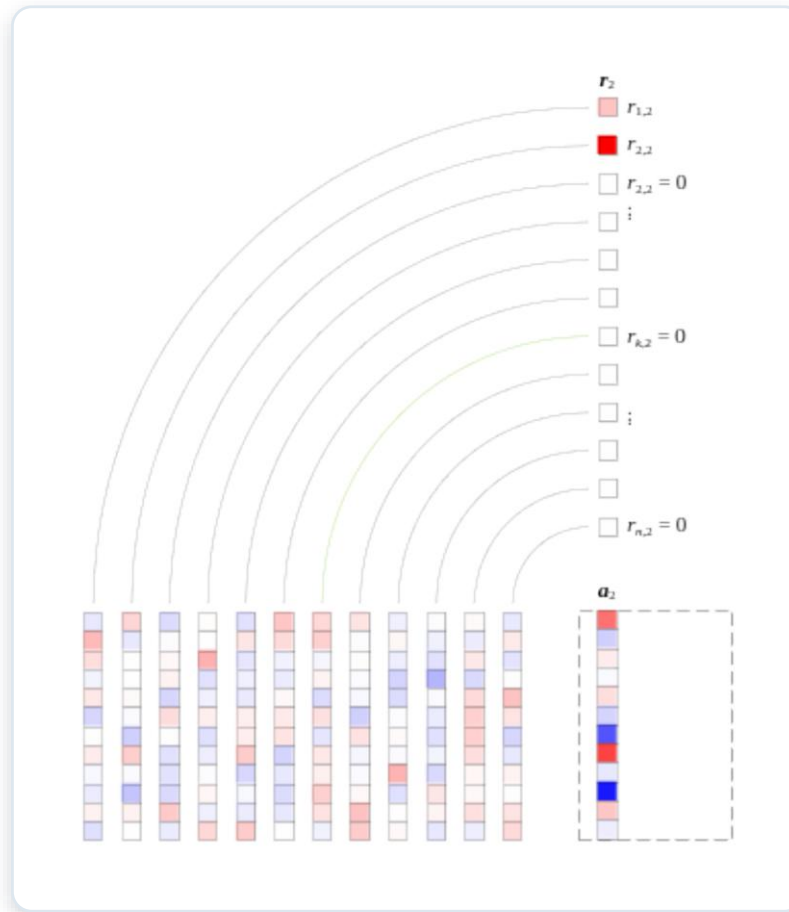
Similarly, A's second column $\mathbf{a}_2$ is written as

$$\mathbf{a}_2 = Q\mathbf{r}_2 = r_{1,2}\mathbf{q}_1 + r_{2,2}\mathbf{q}_2$$

- $\mathbf{a}_2$'s coordinates in orthonormal basis $[\mathbf{q}_1, \dots, \mathbf{q}_n]$ are $(r_{1,2}, r_{2,2}, 0, \dots, 0)$ — only the first two entries are nonzero

Figure 10 · The Matrix Multiplication $\mathbf{a}_2 = \mathbf{Q} \mathbf{r}_2$ for $\mathbf{A}$'s Second Column

- $\mathbf{r}_2$ has only its first two entries nonzero, meaning $\mathbf{a}_2$ lies on the plane $\text{span}(\mathbf{q}_1, \mathbf{q}_2)$



Cutting R horizontally in complete QR decomposition simplifies it into reduced QR decomposition

$$A_{n \times p} = [Q_{n \times p}, Q_{n \times (n-p)}] @ [R_{p \times p}; O_{(n-p) \times p}] = Q_{n \times p} @ R_{p \times p} + Q_{n \times (n-p)} @ O_{(n-p) \times p} = Q_{n \times p} @ R_{p \times p}$$

- The top block is the $p \times p$ upper-triangular $R_{p \times p}$, the bottom block is the $(n-p) \times p$ zero matrix O — block shapes must always match up for multiplication

Figure 11 · Block Matrix Multiplication in Complete QR Decomposition

- Cut R into an upper-triangular block and a zero block; Q splits to match

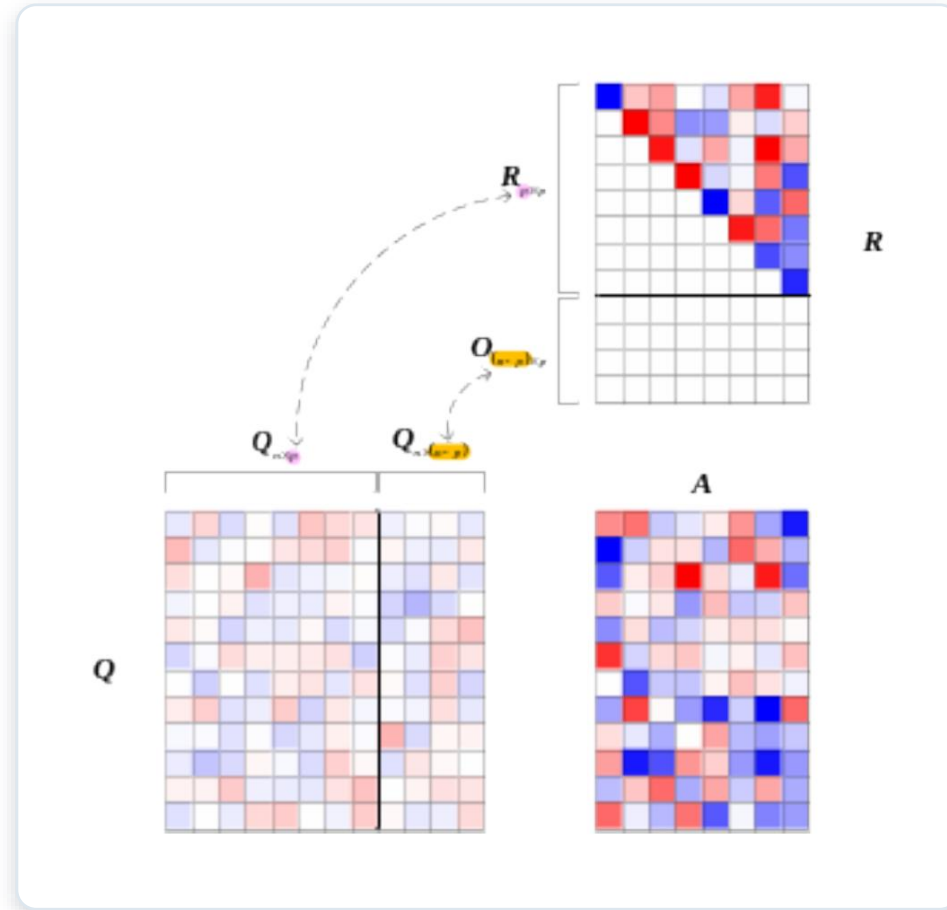


Figure 12 · Adding the Two Matrix Products

- The result is two same-shaped matrices added; the second is all zeros, so it changes nothing

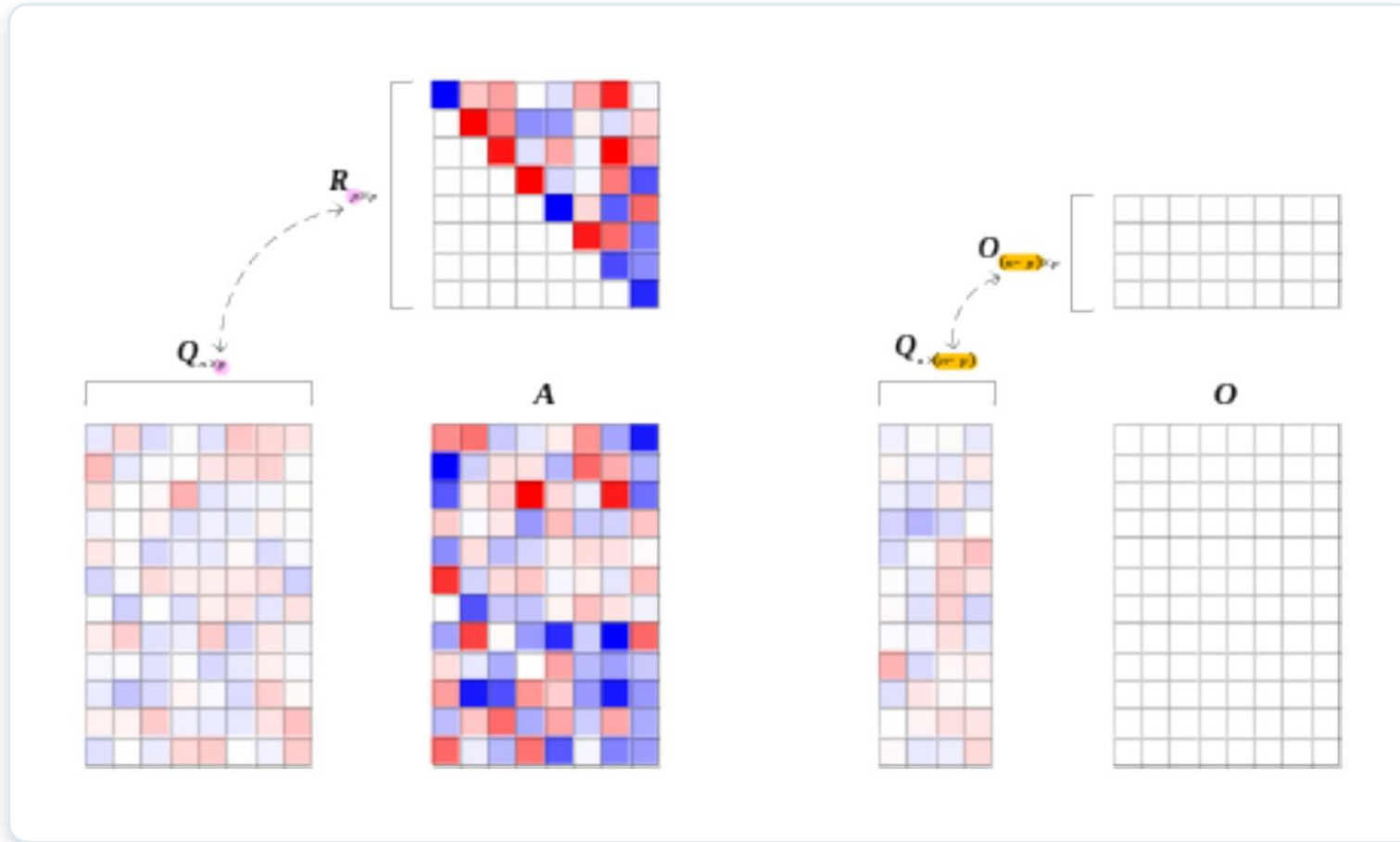
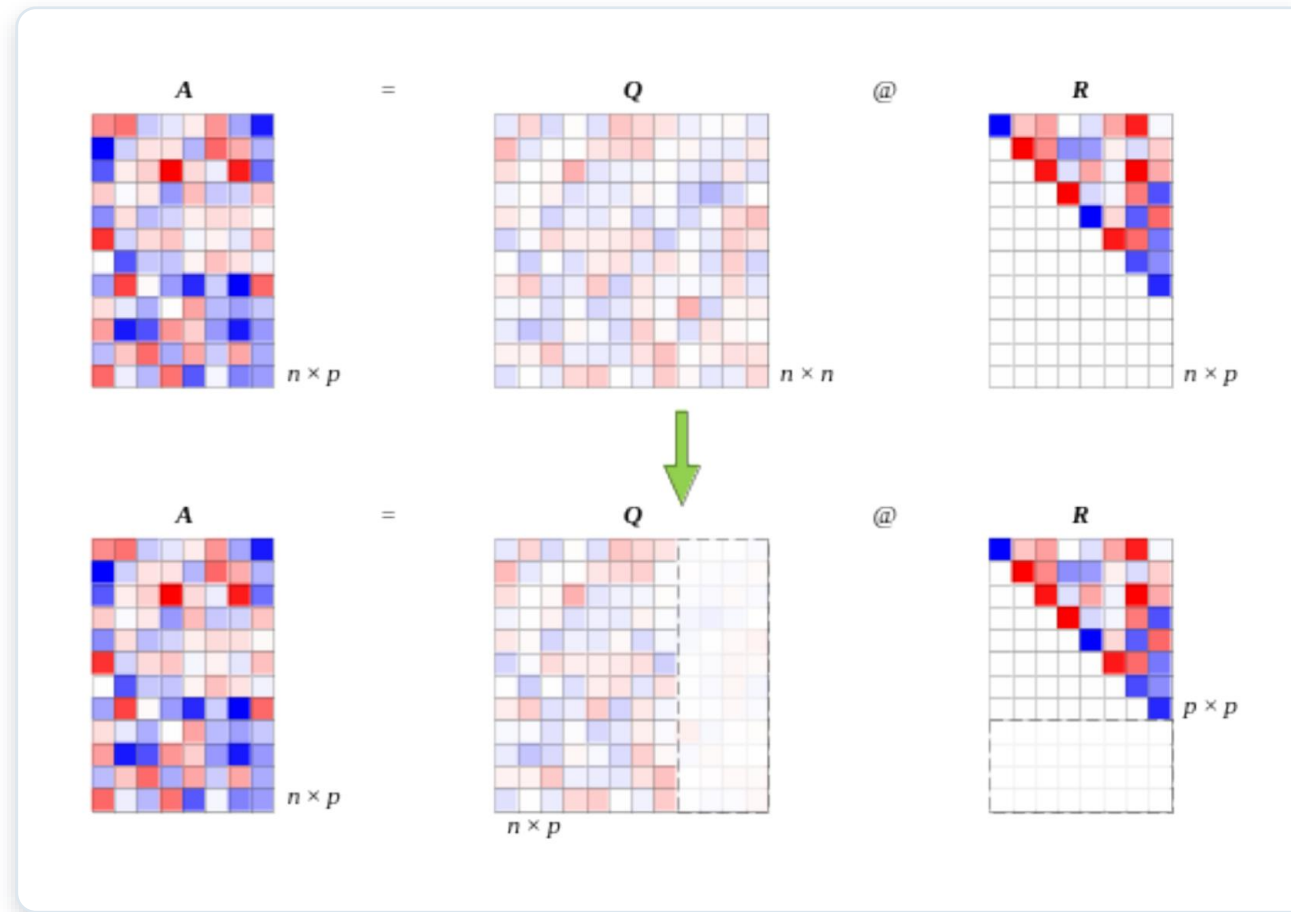


Figure 13 · Simplifying QR Decomposition From Complete to Reduced

- Dropping the zero part, $Q_{n \times p}$ and $R_{p \times p}$ are exactly the reduced QR decomposition



The spaces spanned by $Q_{n \times p}$ and $Q_{n \times (n-p)}$'s columns are orthogonal complements

$$\text{span}(\mathbf{q}_1, \dots, \mathbf{q}_p) \perp \text{span}(\mathbf{q}_{p+1}, \dots, \mathbf{q}_n)$$

- The space spanned by $Q_{n \times p}$'s columns and the space spanned by $Q_{n \times (n-p)}$'s columns are orthogonal complements of each other

Figure 14 · Subspaces That Are Orthogonal Complements

- The two subspaces are perpendicular everywhere, and together they span the entire $\mathbb{R}^n$

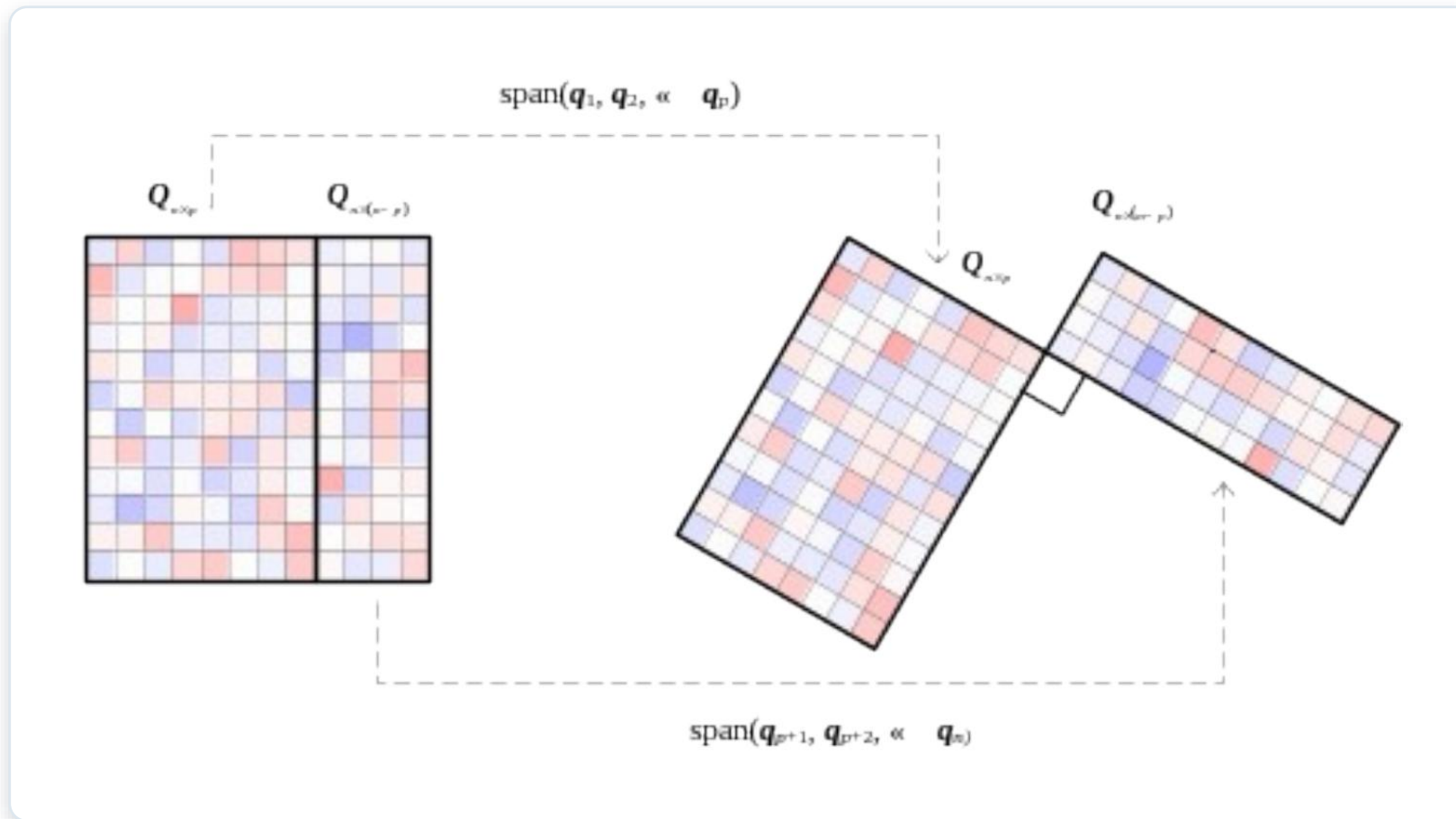
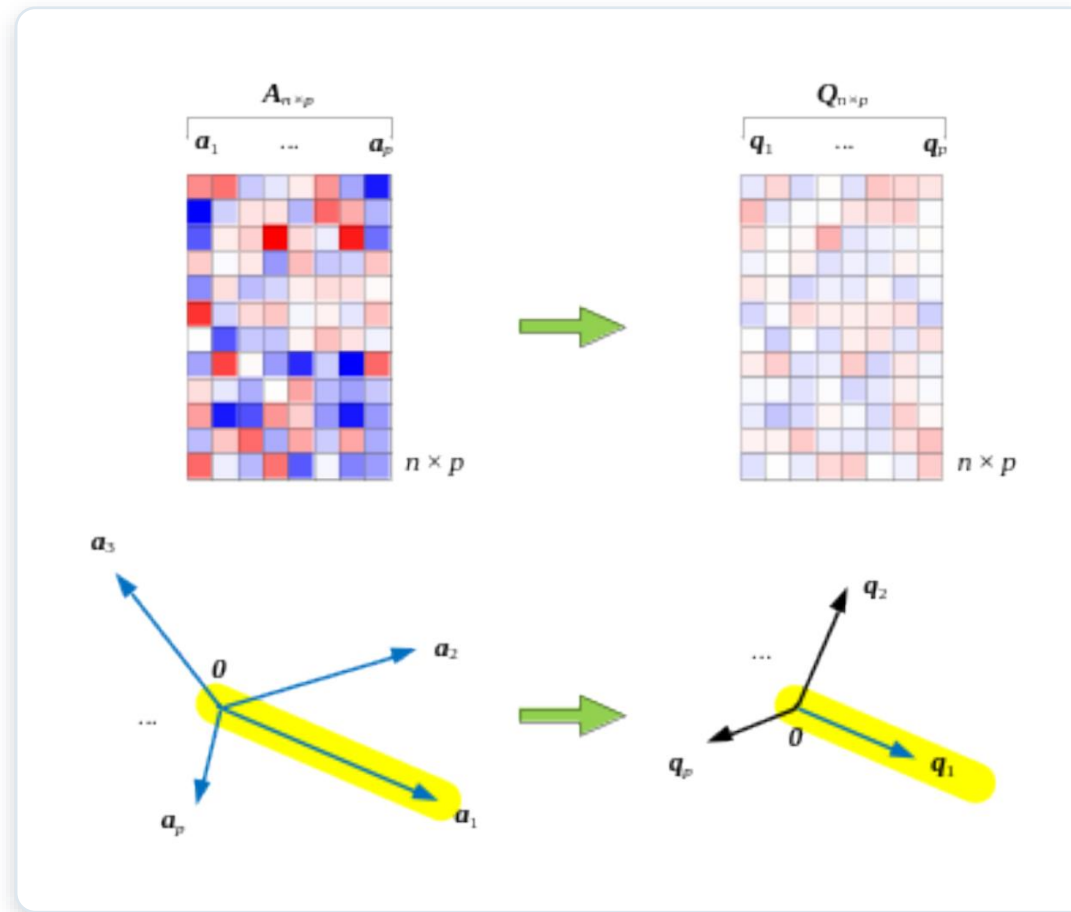


Figure 15 · The Geometric Meaning Behind Reduced QR Decomposition

- Non-orthogonal basis $[a_1, \dots, a_p]$ becomes orthonormal basis $[q_1, \dots, q_p]$



$Q_{n \times p}$ is not square, but its columns are still pairwise orthogonal

$$(Q_{n \times p})^T @ Q_{n \times p} = I_{p \times p}$$

- $Q_{n \times p}$ is a semi-orthogonal matrix — note that $Q_{n \times p}$ is not itself orthogonal, since being orthogonal requires a square matrix

Figure 16 · Semi-Orthogonal Matrix $Q^T @ Q = I$

- $Q_{n \times p}$'s columns are pairwise orthogonal and unit length, but $Q_{n \times p}$ itself is not square

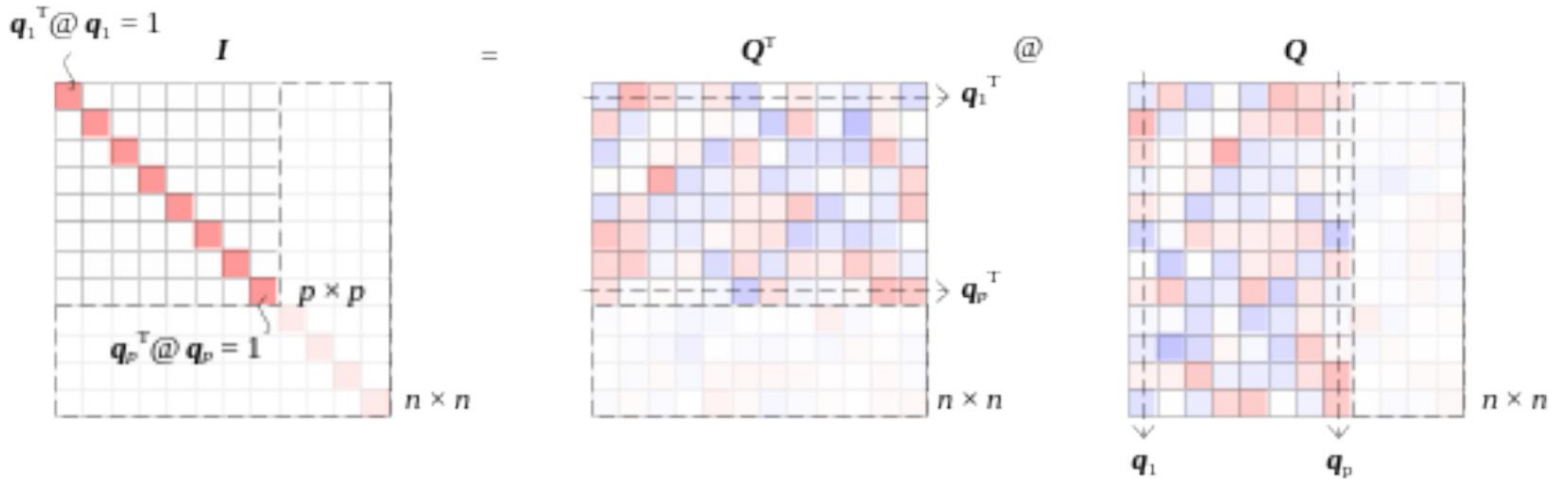
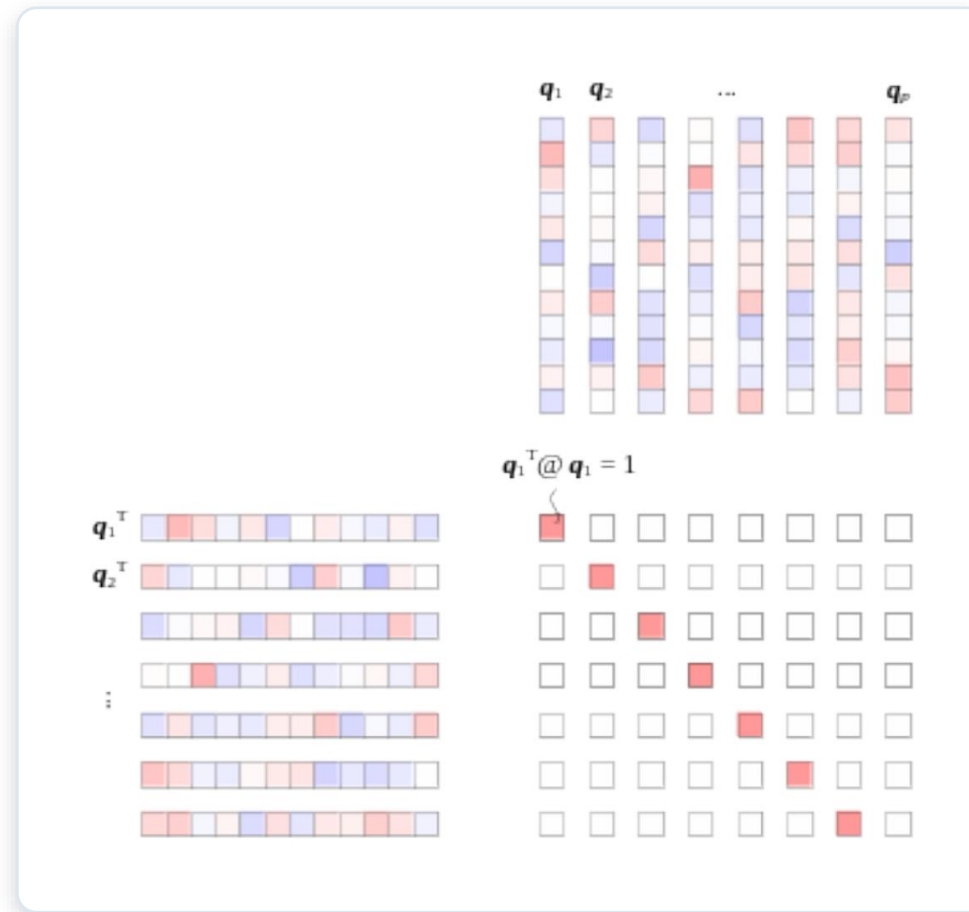


Figure 17 · Semi-Orthogonal Matrix $Q^T @ Q = I$, Matrix Multiplication's First View Expanded

- $q_i^T @ q_j$ is 1 when $i=j$ and 0 when $i \neq j$ — every entry of $I_{p \times p}$ comes from this



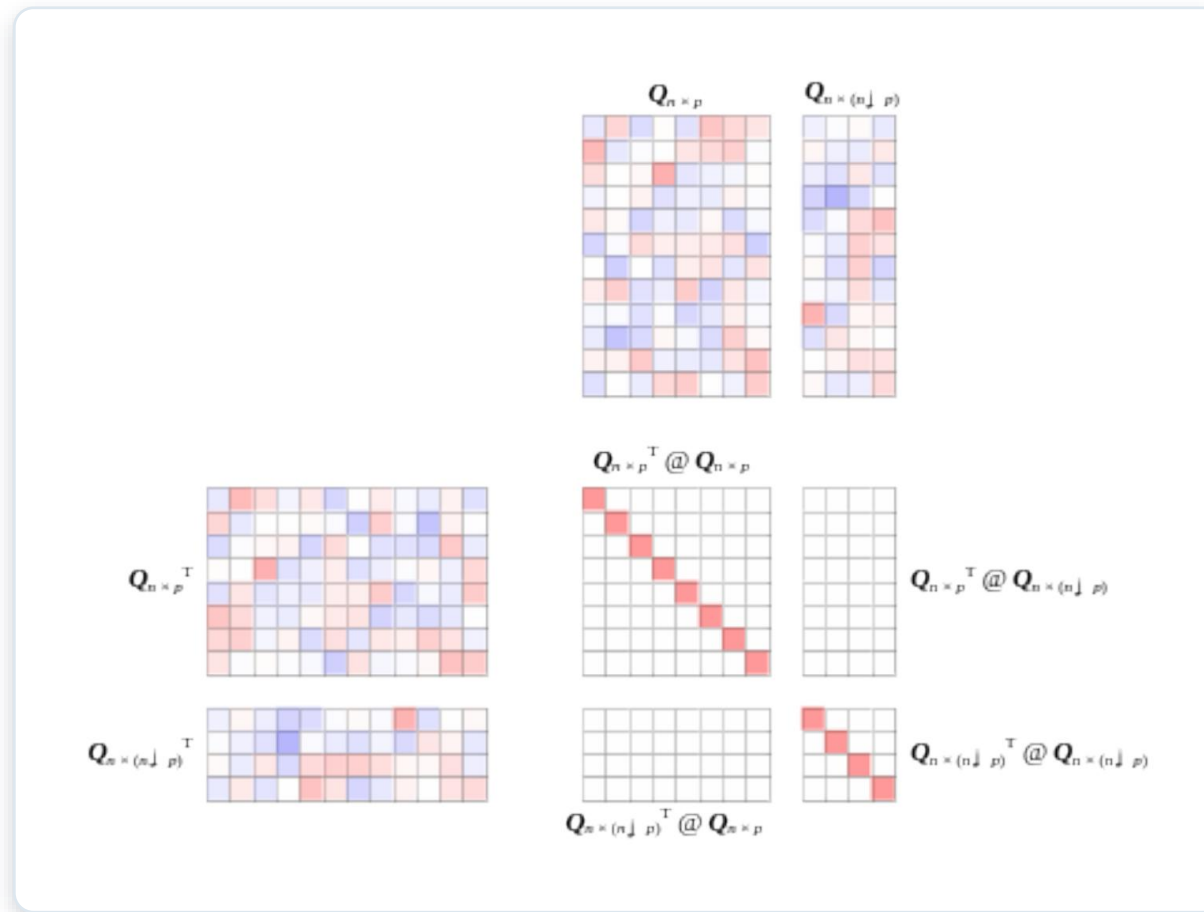
Using block matrix multiplication to get $Q_{n \times p}^T Q_{n \times p}$ from $Q_{n \times n}^T Q_{n \times n}$

$$Q_{n \times n}^T @ Q_{n \times n} = [Q_{n \times p}^T; Q_{n \times (n-p)}^T] @ [Q_{n \times p}, Q_{n \times (n-p)}] = [I_{p \times p}, \mathbf{0}; \mathbf{0}, I_{(n-p) \times (n-p)}]$$

- $Q_{n \times p}$ and $Q_{n \times (n-p)}$ are each semi-orthogonal, so their Gram matrices are identity; since the two are orthogonal to each other, the cross terms are zero matrices

Figure 18 · Getting $Q_{n \times p}^T Q_{n \times p}$ From $Q_{n \times n}^T Q_{n \times n}$

- By blocks: diagonal blocks are identity matrices; off-diagonal blocks are zero



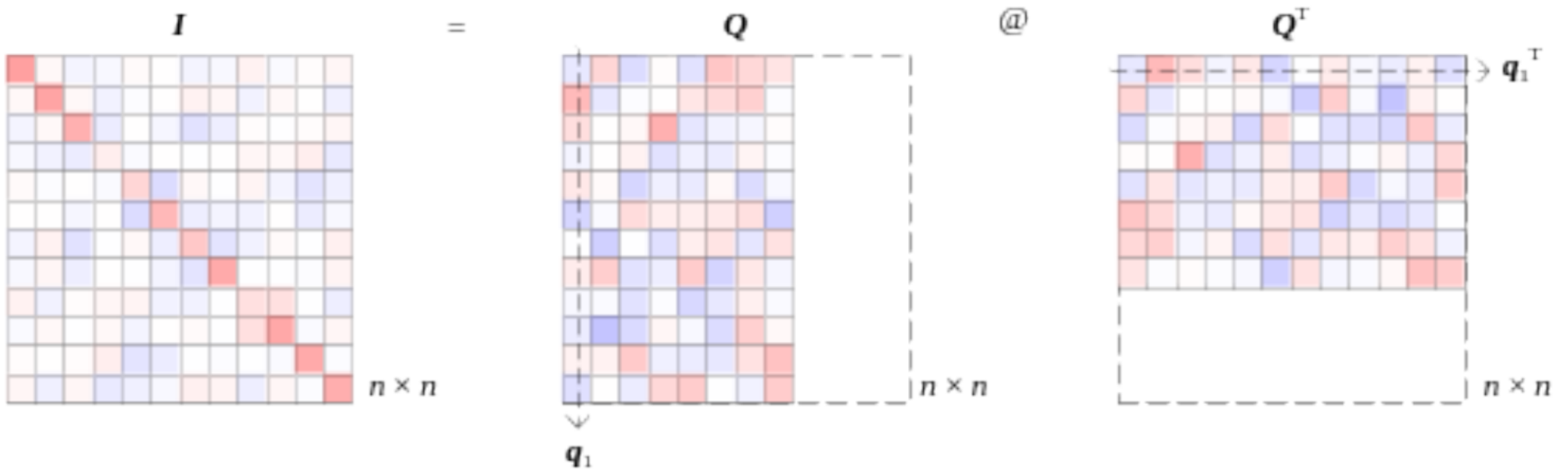
$Q_{n \times p} @ Q_{n \times p}^T$ does not give the identity matrix

$$Q_{n \times p} @ Q_{n \times p}^T \neq I_{n \times n}$$

- When these rank-one matrices are summed, $n-p$ projection matrices are missing, so the result is not the identity

Figure 19 · Semi-Orthogonal $Q@QT=I$'s Result Is Not the Identity Matrix

- Compare Figure 5's complete case — missing $n-p$ columns, so the result isn't the identity



Expanding $Q_{n \times p} Q_{n \times p}^T$ reveals exactly $n-p$ missing directions of projection

$$Q_{n \times p} @ Q_{n \times p}^T = q_1 @ q_1^T + \cdots + q_p @ q_p^T = I - (q_{p+1} @ q_{p+1}^T + \cdots + q_n @ q_n^T)$$

- $q_1 @ q_1^T + \cdots + q_p @ q_p^T$ is the projection matrix onto $\text{span}(q_1, \dots, q_p)$, not the identity matrix

Figure 20 · p Rank-One Matrices Summing to a Square Matrix, Reduced QR Decomposition

- Summing p rank-one matrices covers only $\text{span}(\mathbf{q}_1, \dots, \mathbf{q}_p)$ — n-p directions short



Let's look at matrix A's Gram matrix $A^T A$

$$A^T A = (Q_{n \times p} R_{p \times p})^T (Q_{n \times p} R_{p \times p}) = R_{p \times p}^T Q_{n \times p}^T Q_{n \times p} R_{p \times p} = R_{p \times p}^T R_{p \times p}$$

- Since $Q_{n \times p}^T Q_{n \times p} = I$, the Gram matrix $A^T A$ can be written directly as $R^T R$, with Q dropping out entirely

Figure 21 · Computing Matrix A's Gram Matrix $A^T A$

- $A^T @ A$ is a $p \times p$ symmetric matrix determined by A alone, regardless of decomposition

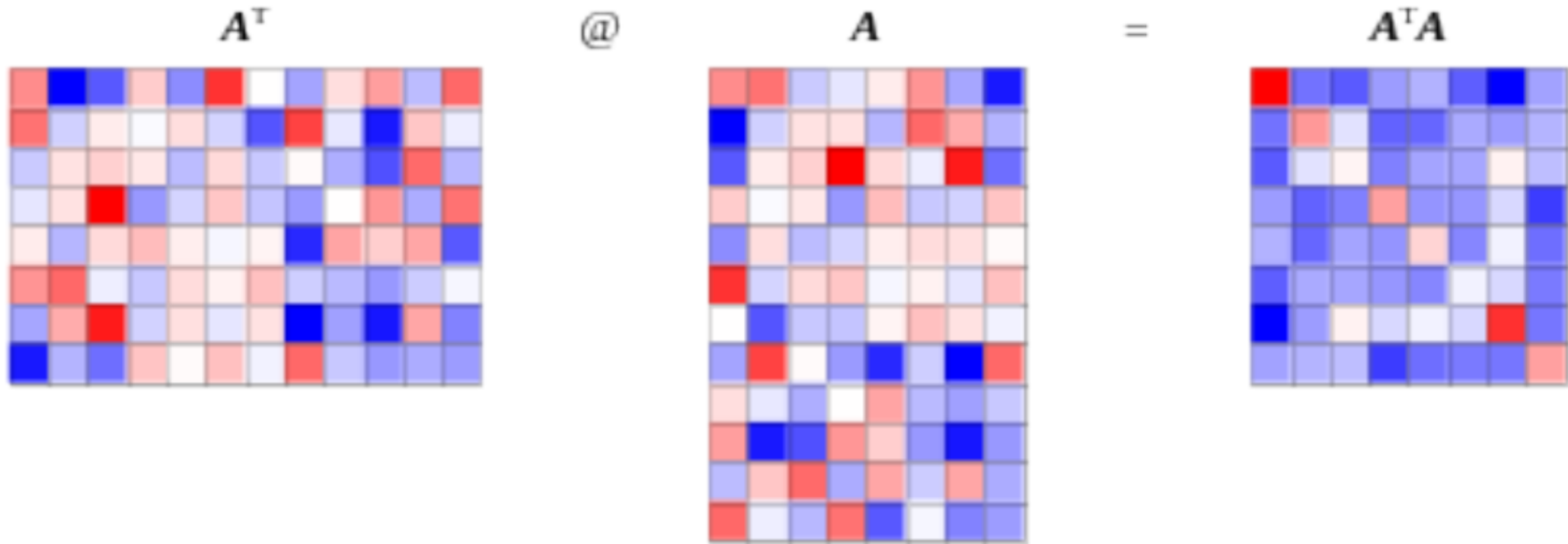
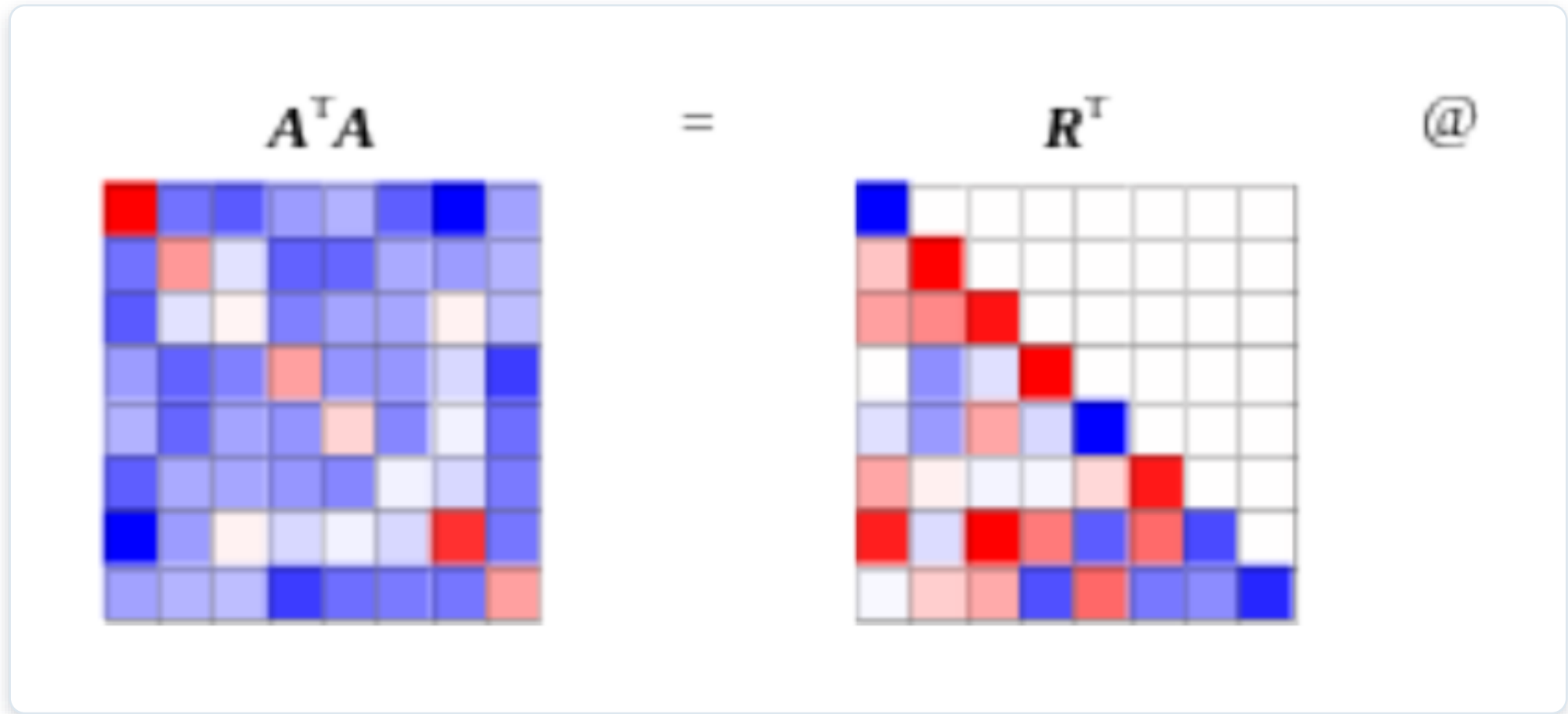


Figure 22 · Decomposing $A^T A$ into $R^T @ R$

- $A^T A$ decomposes into $R^T @ R$ — this is Cholesky decomposition, covered later



If A is a flat matrix (fewer rows than columns), things are different

A (rows < columns, a flat matrix)

- A flat matrix A clearly cannot have full column rank — more columns than the space's dimension forces linear dependence

Figure 23 · QR Decomposition of a Flat Matrix

- $A=Q @ R$ still holds, but zeros can appear on R 's diagonal

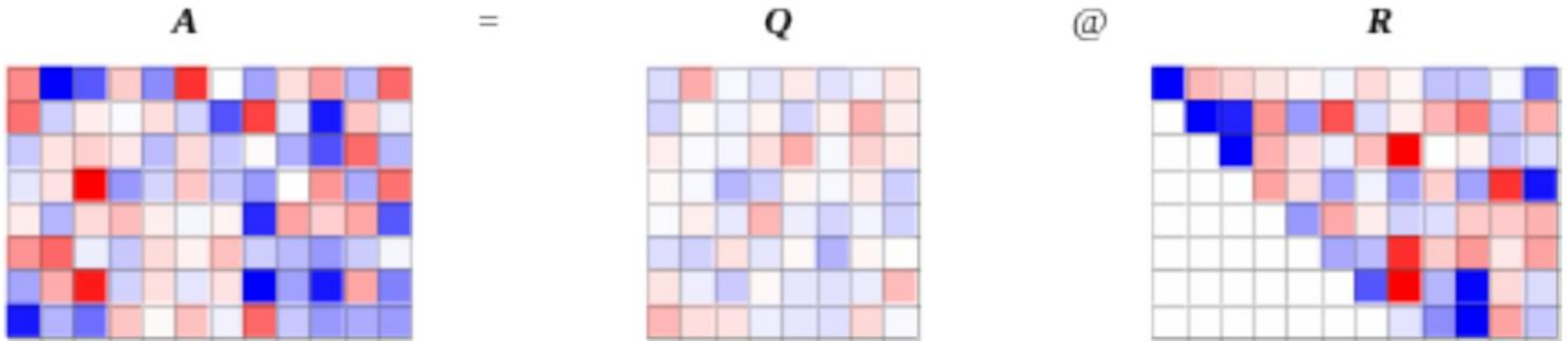


Figure 24 · $QT@Q=I$, Flat Matrix

- Even without full column rank, the resulting Q still satisfies $Q^T@Q=I$

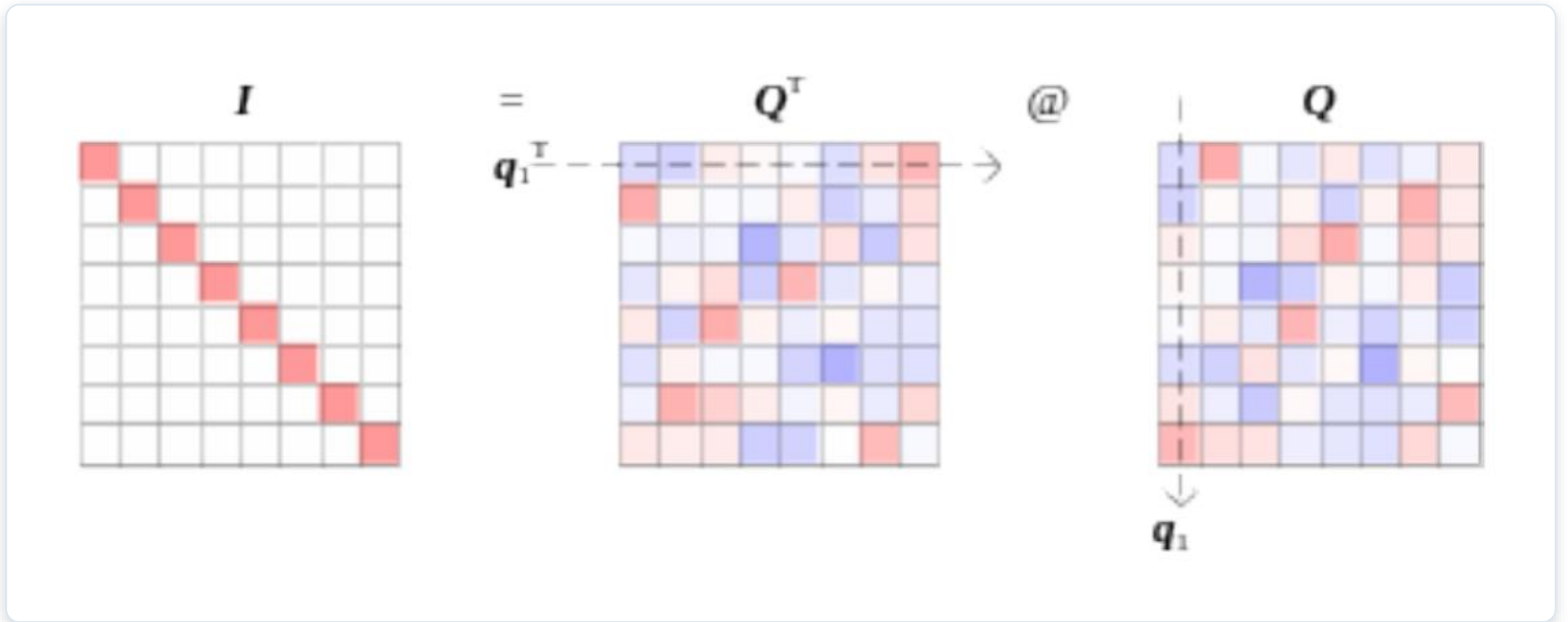
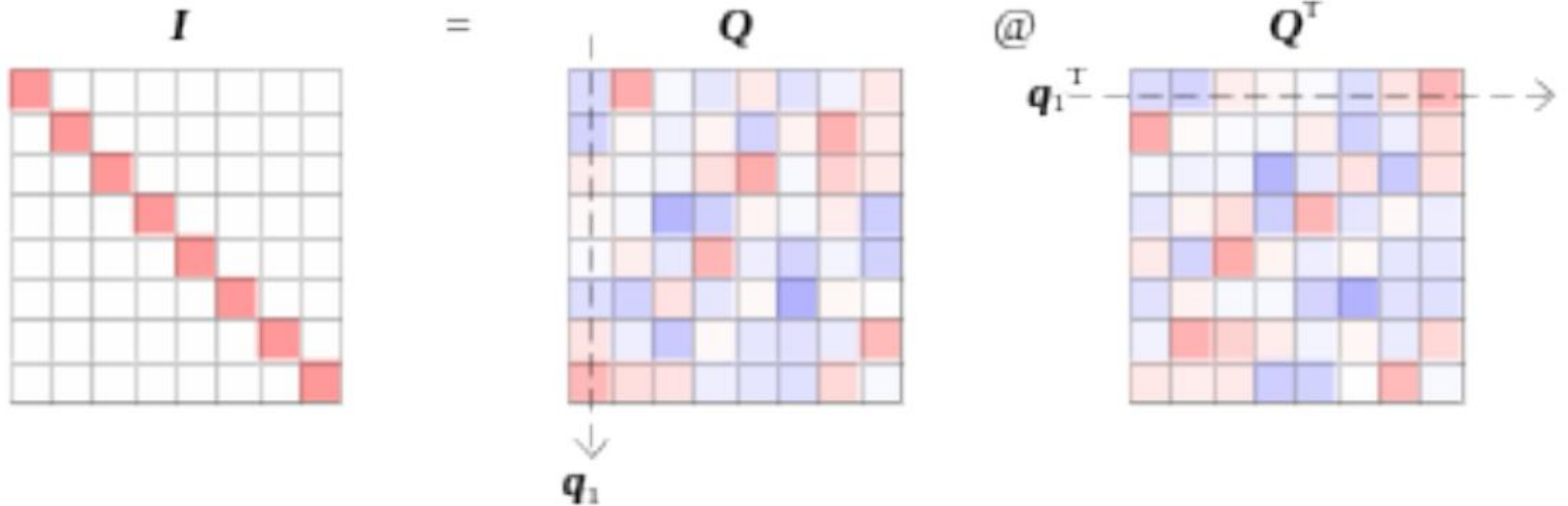


Figure 25 · $Q^T Q = I$, Flat Matrix

- For a flat matrix, Q is square, so $Q^T Q = I$ holds too



What Does QR Decomposition Reveal?

$A = QR$: Q is orthogonal (or semi-orthogonal), R is upper triangular, directly matching Gram–Schmidt orthogonalization

In complete QR decomposition, Q is an $n \times n$ orthogonal matrix; in reduced QR decomposition, Q is semi-orthogonal, matching A 's shape

A semi-orthogonal matrix satisfies $Q^T Q = I$, but $Q Q^T \neq I$ — $n - p$ directions of projection are missing

Each column of R holds the coordinates of A 's corresponding column in the orthogonal basis formed by Q 's columns

Gram matrix $A^T A = R^T R$ previews this book's later Cholesky decomposition

A flat matrix (fewer rows than columns) lacks full column rank; QR decomposition still exists, but zeros can appear on R 's diagonal

Glossary

QR Decomposition QR分解 —— $A=Q @ R$, with Q orthogonal and R upper triangular

Reduced (Thin) QR Decomposition 经济型QR分解 —— Q is semi-orthogonal, with the same shape as A

Orthogonal Complement 正交补 —— the space of all vectors perpendicular to every vector in a given subspace

Gram Matrix 格拉姆矩阵 —— $A^T A$, obtained by multiplying A by its own transpose

Block Matrix Multiplication 分块矩阵乘法 —— multiplying matrices after cutting them into sub-blocks

Complete QR Decomposition 完全型QR分解 —— Q is an $n \times n$ orthogonal matrix, R has the same shape as A

Semi-Orthogonal Matrix 半正交矩阵 —— a matrix satisfying $Q^T Q = I$ but $Q Q^T \neq I$ (fewer columns than rows)

Rank-One Matrix 秩一矩阵 —— a matrix of the form $q @ q^T$, with rank 1

Cholesky Decomposition Cholesky分解 —— decomposing a symmetric positive-definite matrix into $R^T @ R$

Flat (Wide) Matrix 扁平矩阵 —— a matrix with fewer rows than columns, whose columns must be linearly dependent

EXPLORE FURTHER

Further Reading

Recap · 9.4 Gram–Schmidt Orthogonalization

Previous section — step-by-step projection and normalization, building an orthonormal basis

Further Reading · Cholesky Decomposition

Later in this book — decomposing a symmetric positive-definite matrix into $RT^T R$

Further Reading · Householder Transformations

A more numerically stable way to compute QR decomposition

Open-Source Repository

All code and figures for the book: <https://github.com/Visualize-ML>

The Matrix A Used in This Section's Exercises

Matrix	Column 1	Column 2
A (3×2, a counterexample to a flat matrix, full column rank)	$[0,1,1]^T$	$[1,1,0]^T$

A is a 3×2 matrix with more rows than columns and full column rank; compute A's complete QR decomposition, where Q is a 3×3 orthogonal matrix and R is 3×2.

Q1 – Q9

Q1 Write Python code to compute the complete QR decomposition of the matrix A in the table above.

Q3 For matrix A above, verify by hand whether A has full column rank (whether its columns are linearly independent).

Q5 Use `numpy.linalg.qr(A, mode='reduced')` to compute A's reduced QR decomposition, and compare Q's and R's shapes with the complete mode.

Q7 Verify the reduced-case Q satisfies $QTQ=I$ but $QQT \neq I$, and compute what QQT actually equals.

Q9 Use `numpy.linalg.cholesky(A.T @ A)` to compute a Cholesky decomposition, and compare it against the R from QR decomposition.

Q2 Learn how the Householder algorithm is used to compute a QR decomposition.

Q4 Use `numpy.linalg.qr(A, mode='complete')` to compute A's complete QR decomposition, and compare with your hand calculation.

Q6 Verify the complete-case Q satisfies $QTQ=QQT=I$ (check both identities separately).

Q8 Compute matrix A's Gram matrix ATA , and use the reduced QR decomposition to verify $ATA=RTR$.

Q10 – Q17

Q10 Write a Python function `verify_qr(A)` that computes A's QR decomposition and checks whether $A=Q @ R$ holds (within $1e-10$ numerical tolerance).

Q12 Prove: if A has full column rank, then none of R's diagonal entries can be zero.

Q14 Explore: QR decomposition results are not unique — construct an example showing the same A can correspond to two different (Q,R) decompositions (hint: flip the sign of R's diagonal).

Q16 Explore: why are QR decomposition and Gram–Schmidt orthogonalization essentially the same thing? How do their computation orders differ?

Q11 Explore: swap the two columns of matrix A and redo QR decomposition — observe how the sign of R's diagonal entries changes.

Q13 Write a Python function `is_upper_triangular(R)` and use it to verify this section's example.

Q15 Using Gram–Schmidt orthogonalization (§9.4's method), compute matrix A's Q by hand, and compare against `numpy.linalg.qr`'s result.

Q17 Write a Python function `flat_matrix_qr(A)` that computes the QR decomposition of a matrix with fewer rows than columns, and observe Q's and R's shapes.

Q18 – Q25

Q18 For the flat matrix from Q17, verify whether $QTQ=I$ still holds even though the columns are not linearly independent (hint: a flat matrix's columns must be linearly dependent).

Q20 For the 5×3 random matrix in Q19, verify that $q1@q1T+q2@q2T+q3@q3T$ equals $I-(q4@q4T+q5@q5T)$.

Q22 Write Python code using matplotlib to visualize how a set of vectors changes before and after QR decomposition (a 3D example).

Q24 Compute the determinant of R from matrix A's QR decomposition (using a 3×2 square sub-case of A), and compare it against the square root of $\det(ATA)$ to verify the relationship.

Q19 Using a randomly generated 5×3 matrix, verify whether the subspaces spanned by $Q_{n \times p}$ and $Q_{n \times (n-p)}$ in the complete QR decomposition are orthogonal complements.

Q21 Explore: use QR decomposition to solve a linear system $Ax=b$ (A square and invertible), and compare its numerical stability with directly inverting A.

Q23 Explore: SVD, Gram–Schmidt orthogonalization, and QR decomposition can all produce orthogonal matrices — research and compare where each is best applied and how numerically stable each is.

Q25 Preview thought: the Gram matrix ATA 's Cholesky decomposition $ATA=R^TR$ — how does it relate to the R from this section's QR decomposition? Explain your reasoning.

10 Prompts to Explore With AI

1 Explain the difference between complete and reduced QR decomposition, and when each is used.

2 Explain why a semi-orthogonal matrix Q satisfies $QTQ=I$ but not $QQT=I$.

3 Use Python to demonstrate the connection between QR decomposition and Gram–Schmidt orthogonalization, and verify whether the results agree.

4 Explain the geometric meaning of each column of R , and why its diagonal entries can be positive or negative.

5 Explain why QR decomposition results are not unique, and how requiring R 's diagonal entries to be positive makes the decomposition unique.

10 Prompts to Explore With AI

6 Explain the relationship between Gram matrix $ATA=RTR$ and Cholesky decomposition.

7 Explain why the subspaces spanned by $Q_{n \times p}$ and $Q_{n \times (n-p)}$ are orthogonal complements.

8 Explain why, for a flat matrix (fewer rows than columns), Q 's columns cannot be guaranteed linearly independent when doing QR decomposition.

9 Explain how Householder transformations are used to compute a numerically stable QR decomposition, and their advantage over Gram–Schmidt orthogonalization.

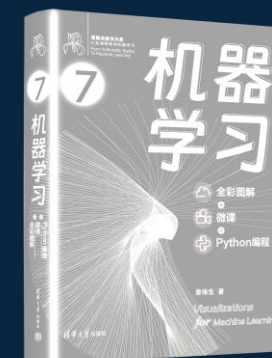
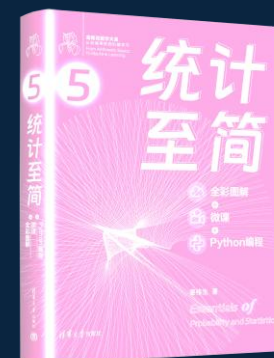
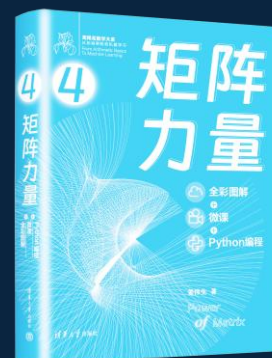
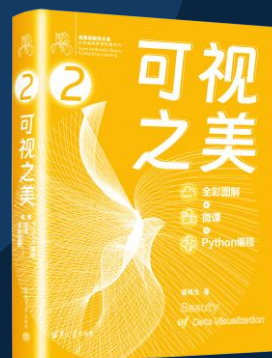
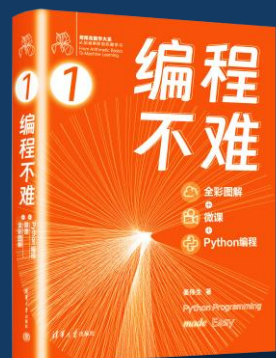
10 Explain the role QR decomposition can play in solving linear systems and least-squares problems.

数学

数学基础

线性代数

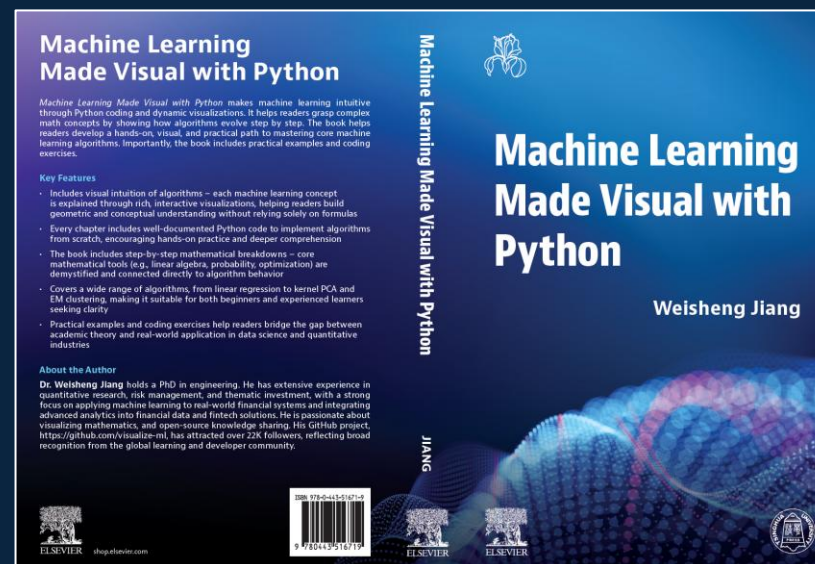
概率统计



Python编程

数据可视化

工具



数据分析
图和网络

回归、降维
分类、聚类

实践